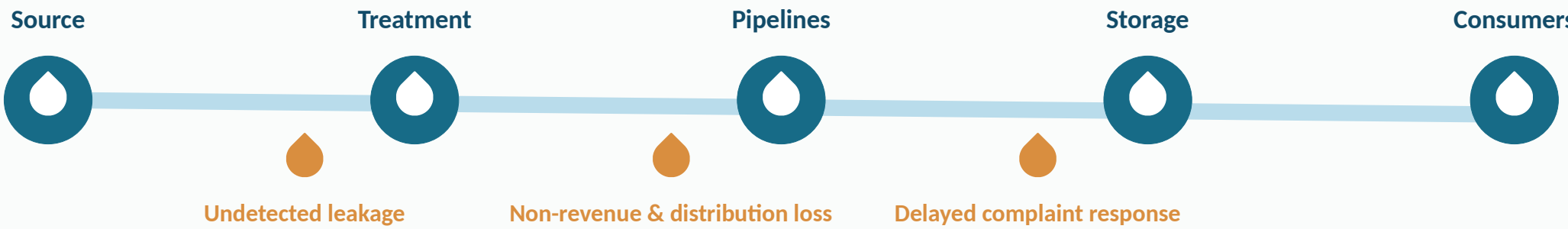
Water Is Lost Before We Notice



Where water and visibility are lost

- Undetected pipeline leakage
- Non-revenue water losses
- Delayed complaint response
- Irregular supply, unbalanced demand
- Fragmented infrastructure data
- Reactive, not proactive, maintenance

Who is affected

- Residents and households
- Municipal authorities, utility operators
- Maintenance teams
- Hospitals and essential services
- Industrial and institutional users
- Local communities

Why sensors alone fall short

- Limited coverage and blind spots
- Sensor failure hides problems
- Misses uninstrumented streets
- Data may miss citizen experience
- Citizen reports fill the gaps

"What if the infrastructure cannot detect a problem, but the community already knows it exists?"

AI Reasoning and Agent Architecture

INPUT	AI / AGENTS	TOOLS	DATA / KNOWLEDGE	DECISIONS	HUMAN CONTROL	OUTPUT
• Sensor readings • Citizen reports • Weather • Historical and infrastructure data	• Anomaly Detection • Root-Cause Assessment • Demand Forecasting • Shortage Assessment • Citizen Intelligence • Decision Support	• Python ML models • NLP, geospatial analysis • Database queries • OR-Tools optimisation • Water-balance calculations	• PostgreSQL time series • Historical incidents • Verified maintenance records • Citizen report history	• Suspected anomalies • Possible root causes • Demand and shortage • Prioritised repairs • Resource allocation	• Review evidence • Verify critical incidents • Approve or reject • Record decisions	• Investigation priorities • Repair plans • Distribution plans • Alerts and impact reports

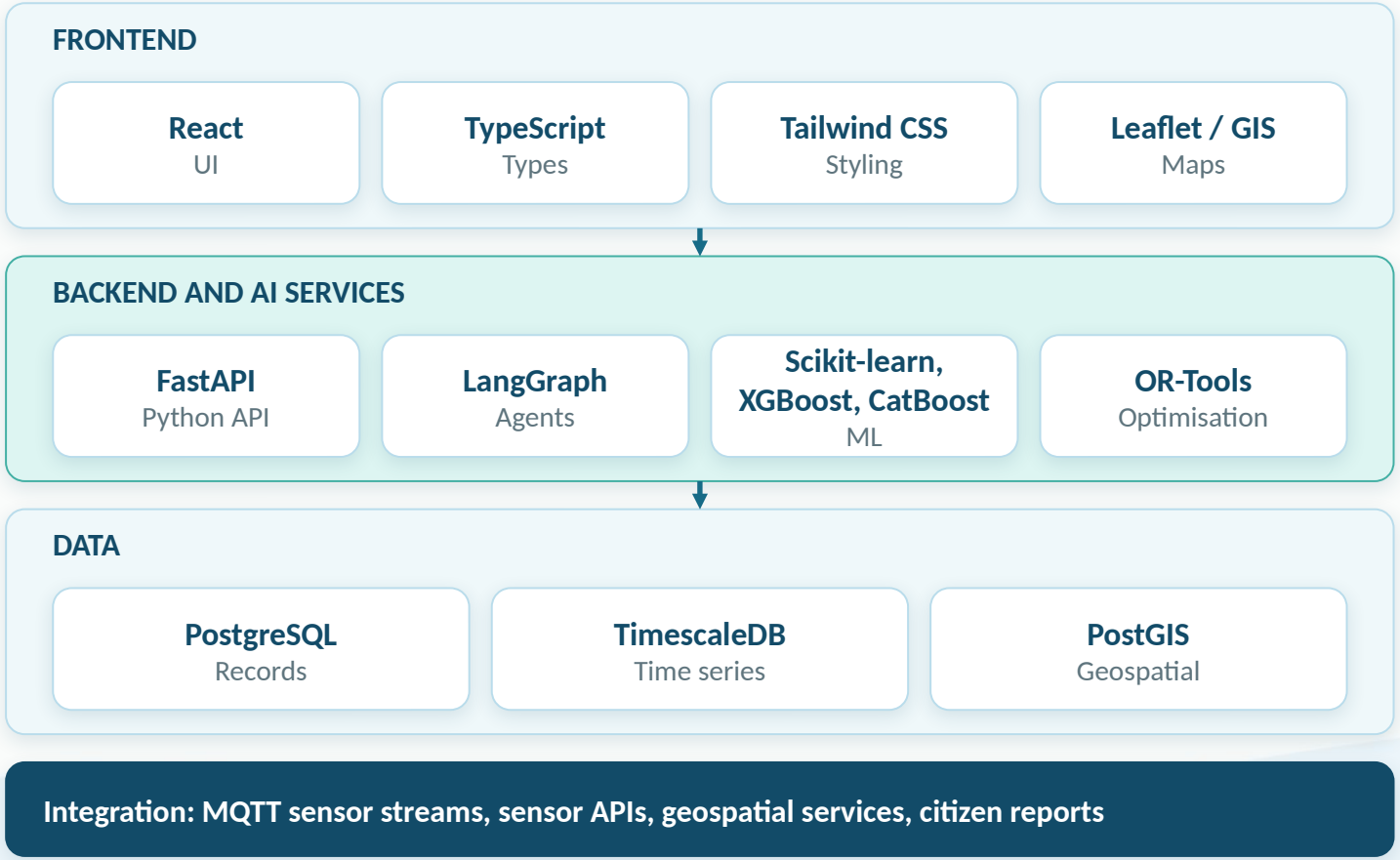
AI-GENERATED ANALYSIS AND RECOMMENDATIONS

HUMAN-AUTHORISED ACTIONS

1 Detect → **2 Understand** → **3 Predict** → **4 Optimise** → **5 Verify**

The LLM explains and orchestrates. Numbers come from ML models, OR-Tools and water-balance calculations. People authorise every action.

Technical Architecture and Prototype



Models and tools

- **One-Class SVM**
  Anomaly detection. Score is not a leak probability.
- **XGBoost Classifier**
  Root cause: leak, demand, meter or sensor fault.
- **CatBoost Regressor**
  Demand forecasting from history, time, weather.
- **Water balance**
  Next = Current + Inflow − Outflow − Losses
- **Google OR-Tools**
  Allocation under capacity and essential-service limits.
- **Decision engine**
  Combines outputs, evidence, constraints, priorities.

Signals → validate → anomaly → root cause → forecast → evidence → recommendation → human approval → verification

Evidence, Evaluation and Responsible AI

Conclusion: possible pipeline leakage (unverified, awaiting human review)

Claim	Source	Operation	Limitation	Status
3 related reports, one street	Citizen reports	Spatial clustering	May be duplicates	**Unverified lead**
Unusual flow reading	Flow sensor	One-Class SVM	Not a leak probability	**Model assessment**
Pressure below baseline	Pressure sensor	Baseline check	Verify sensor health	**Observed fact**

Illustrative example, not real data. A confidence score is not proof.

EVALUATION TARGETS. NO RESULTS CLAIMED.

ML
Anomaly: precision, recall, FPR. Classifier: F1, confusion matrix. Forecast: MAE, RMSE

Citizen intelligence
Complaint F1, duplicate precision, hotspot quality, report-to-investigation time

Operations
Investigation time, estimated vs verified recovery, essential-demand satisfaction

Responsible AI
Citation correctness, unsupported-claim and abstention rates, human overrides

1. **Privacy**
   Collect only what is needed; pseudonymise; retention limits
2. **Security**
   Authenticated, role-based access; secure APIs; audit logs
3. **Hallucination control**
   Grounded in records; evidence shown; abstain when data is thin
4. **Human approval**
   AI recommends, people decide; decisions logged
5. **Failure and fairness**
   Flag stale sensors; silence is not proof; report volume never sets priority alone; equity constraints

Adaptive Resilience and Impact

SIMULATION

A 40% supply drop

- Water balance recalculated
- Essential demand protected first
- Distribution plan revised
- Human approval

SIMULATION

B Sensor failure

- Stale readings detected
- Source reliability lowered
- Reports and history weighed in
- Lower confidence; verify on site

SIMULATION

C Demand surge

- Forecast updated
- Shortage risk assessed
- Options recalculated
- Conservation actions suggested

HYPOTHETICAL IMPACT: ASSUMPTIONS ARE EDITABLE, NOT CLAIMS OF ACTUAL SAVINGS

10,000 kL/day supply × 20% assumed loss × 25% assumed recoverable share × ₹20/kL assumed value

Daily loss	Potential daily recovery	Annual potential recovery	Illustrative annual value
2,000 kL	500 kL	182,500 kL	₹36.5 lakh

Actual recovery depends on field verification, repair effectiveness and measurement. Benefits: less freshwater wastage, more reliable essential services, better shortage resilience, support for underserved communities.

Our Unique Value Proposition

From Community Signals to Intelligent Water Resilience.

1 Citizen-powered intelligence

Community reports supplement sensor networks and extend visibility.

2 Evidence-grounded AI

Sources are correlated before any conclusion is shown.

3 Human-controlled decisions

The system recommends and simulates; authorised people approve.

Dimension	Basic sensor-centric workflow	NEERARAN
Data sources	Mainly infrastructure sensors	**Sensors + citizen reports + history + context**
Analysis	Isolated alerts	**Correlated multi-source assessment**
Planning	Manual or fragmented	**Constraint-aware optimisation support**
Accountability	Limited traceability	**Evidence and approval records**

NEERARAN does not simply monitor water infrastructure. It connects, reasons, predicts and supports action.

COMMERCIAL VIABILITY

Business Model

Target customers

- Municipal water authorities
- Water utilities
- Industrial campuses
- Educational institutions
- Large residential communities
- Infrastructure operators

Business value

- Reduce avoidable water loss
- Faster suspected-leak investigation
- Prioritised maintenance spend
- Better distribution planning
- Lower emergency repair costs
- Less manual monitoring
- Operational accountability

Proposed revenue model

- SaaS subscription
- Tiers by zones, assets or data volume
- Deployment and integration fees
- Optional analytics and optimisation modules
- Support and maintenance contracts

A proposed commercial approach, not market data. No customer contracts or revenue figures are claimed.

NEERARAN | Beyond Sensors. Beyond Monitoring. Intelligence for Every Drop.